An Overview of Data Assimilation: From Smoothing to Filtering Algorithms

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Abstract

This review aims at presenting the general ideas underlying Data Assimilation by introducing the notions that form the basis of smoothing and filtering algorithms in both discrete and continuous time. This review is exclusively based on the work of K. Law, A. Stuart & K. Zygalakis in [7].

1 Data Assimilation in Discrete-Time

Let $\Psi \in C(\mathbb{R}^l, \mathbb{R}^l)$ and set the discrete-time dynamical model

$$v_{j+1} = \Psi(v_j) + \xi_j \tag{1}$$

for $v_0 \sim \mathcal{N}(m_0, C_0)$ and $\xi = \{\xi_j\}_{j\in\mathbb{Z}^+}$ is an i.i.d. sequence with $\xi_0 \sim \mathcal{N}(0,\Sigma)$ such that $\Sigma \succ 0$. The set $\{v_j\}_{j\in\mathbb{Z}^+}$ is herby called the **signal** and the probability density function (*pdf*) of v quantifies the uncertainty in predictions arising in the stochastic dynamical model. We then observe a (possibly non-linear) function of the signal where the **observations** are noisy:

$$y_{j+1} = h(v_{j+1}) + \eta_{j+1} \tag{2}$$

for $h \in C(\mathbb{R}^n, \mathbb{R}^m)$ and $\eta = \{\eta_j\}_{j\in\mathbb{Z}^+}$ is an i.i.d. sequence independent of (v_0, ξ) with $\eta_1 \sim \mathcal{N}(0,\Gamma)$ such that $\Gamma \succ 0$. Here, h is the **observation operator** and the objective of **data assimilation** is to determine information about the signal v given data y. Note that Ψ could for instance be the solution to an ordinary differential equation (ODE).

1.1 Smoothing

The **smoothing distribution** is the random variable $v|y$ and contains all the probabilistic information about the signal defined on a discrete-time interval $\mathbb{J}_0 = \{0,\dots,J\}$ given some data y generated on a discrete-time interval $\mathbb{J} = \{1,\dots,J\}$. The idea to write the smoothing distribution will be to use Bayes' formula. In [7, pp. 34-35], it is shown that the **prior** distribution is given by

$$\mathbb{P}(v) \propto \exp(-J(v)) \tag{3}$$

where $J(v)$ is given by:

$$J(v) := \frac{1}{2}\left|v_0 - m_0\right|^2_{C_0} + \frac{1}{2}\sum_{j=0}^{J-1}\left|v_{j+1} - \Psi(v_j)\right|^2_{\Sigma}.$$

We remark that the prior is in general not a Gaussian. Moreover, it is also shown that the **likelihood** is of the form

$$\mathbb{P}(y|v) \propto \exp(-\Phi(v;y)) \tag{4}$$

where $\Phi(v;y)$ is given by:

$$\Phi(v;y) := \frac{1}{2}\sum_{j=0}^{J-1} \left|y_{j+1} - h(v_{j+1})\right|_{\Gamma}^2. \tag{5}$$

We remark that, under our setup, the likelihood is Gaussian. Now, by Bayes' formula, the **posterior** distribution corresponding to the smoothing distribution is a probability measure μ on $\mathbb{R}^{|\mathbb{J}_0|\times n}$ with a *pdf* given by

$$\mathbb{P}(v|y) \propto \mathbb{P}(y|v)\mathbb{P}(v) \propto \exp(-I(v;y)) \tag{6}$$

where $I(v;y) := J(v) + \Phi(v;y)$. Note that the posterior is in general not Gaussian unless Ψ and h are linear.

1.2 Filtering

Recall that smoothing is the idea of finding the distribution of the conditioning $v_j|y_k$ where generally $k > j$. This corresponds to an *offline* scenario in which we are able to compute this conditioning after the generation of some data. On the other hand, the **filtering distribution** aims to determine the random variable $v|y$ via a conditioning $v_j|Y_j$ where $Y_j = \{y_l\}_{l=1}^{j}$. In particular, filtering is the procedure of sequentially updating the *pdf* $\mathbb{P}(v_j|Y_j)$ as the index j is incremented. Note now that this is an *online* scenario in which we want to determine the state of the signal in real time. Indeed, the data Y_j never surpasses the state of the signal v_j. To determine the update of the *pdf*, we will again use Bayes' formula. The idea is to first compute a **prediction** providing a mapping $\mathbb{P}(v_j|Y_j) \mapsto \mathbb{P}(v_{j+1}|Y_j)$, and then compute an **analysis** providing the mapping $\mathbb{P}(v_{j+1}|Y_j) \mapsto \mathbb{P}(v_{j+1}|Y_{j+1})$. Note that, since Y_j contains noisy and indirect information about v_j and cannot improve upon perfect knowledge of v_j, we have that $\mathbb{P}(v_{j+1}|Y_j, v_j) = \mathbb{P}(v_{j+1}|v_j)$. Thus, by the definition of a conditional probability:

$$\mathbb{P}(v_{j+1}|Y_j) = \int_{\mathbb{R}^n} \mathbb{P}(v_{j+1}|Y_j, v_j)\mathbb{P}(v_j|Y_j)\,\mathrm{d}v_j = \int_{\mathbb{R}^n} \mathbb{P}(v_{j+1}|v_j)\mathbb{P}(v_j|Y_j)\,\mathrm{d}v_j. \tag{7}$$

Moreover, since Y_j contains noisy and indirect information about v_j and cannot improve upon perfect knowledge of v_{j+1}, we have that $\mathbb{P}(y_{j+1}|v_{j+1}, Y_j) = \mathbb{P}(y_{j+1}|v_{j+1})$. Thus from Bayes' formula:

$$\mathbb{P}(v_{j+1}|Y_{j+1}) = \mathbb{P}(v_{j+1}|Y_j, y_{j+1}) = \frac{\mathbb{P}(y_{j+1}|v_{j+1}, Y_j)\mathbb{P}(v_{j+1}|Y_j)}{\mathbb{P}(y_{j+1}|Y_j)} = \frac{\mathbb{P}(y_{j+1}|v_{j+1})\mathbb{P}(v_{j+1}|Y_j)}{\mathbb{P}(y_{j+1}|Y_j)} \tag{8}$$

where Eq. (2) determines $\mathbb{P}(y_{j+1}|v_{j+1})$. We note the caveat that we do not have easily usable closed-form expressions for $\mathbb{P}(v_j|Y_j)$, and thus we require algorithms to provide approximations. Nonetheless, Eqs. (7) and (8) fully determine a **filtering update** for an algorithm.

2 Discrete-Time Smoothing Algorithms

We saw that the posterior distribution is in general a hard quantity to compute, as a result we must estimate it. A specific case where we can determine the posterior distribution exactly is the **Kalman Smoother**. This corresponds to the case where Ψ and h are linear. Thus we set $\Psi(v) = Mv$ and $h(v) = Hv$ such that $M \in \mathbb{R}^{n\times n}$ and $H \in \mathbb{R}^{m\times n}$. It is shown in [7, Theorem 3.1] that the posterior smoothing distribution is Gaussian $\mu = \mathcal{N}(m, C)$ for $C = L^{-1}$ is the inverse of a precision matrix. However, for harder probability distributions, we must consider some approximations. One solution

is to consider Monte Carlo sampling. Indeed, for a measure ν, we generate N samples $\{v^{(j)}\}_{j=1}^N$ (approximately) drawn from ν via **Markov Chain Monte Carlo (MCMC)** algorithms. This will allow us to conclude that:

$$\nu^N := \frac{1}{N}\sum_{j=1}^{N}\delta_{v^{(j)}} \simeq \nu. \tag{9}$$

We note that the resulting measure ν^N converges to the true measure ν as $N \to \infty$. Common choices of algorithms for MCMC are Metropolis-Hastings, Random Walk Metropolis, pCN sampler, only to name a few. Another possibility to approximate the posterior distribution is given by **variational methods**. Alternative names are *4DVAR* in Atmospheric Sciences or *Maximum A Posteriori (MAP)* estimation in Statistics, for which we must solve the optimization problem

$$v^* = \underset{v\in\mathbb{R}^{|\mathbb{J}_0|\times n}}{\arg\max}\ \mathbb{P}(v|y). \tag{10}$$

Alternatively, a generalization of 4DVAR is given by *w4DVAR* (weak 4DVAR) where the minimization problem becomes

$$v^* = \underset{v\in\mathbb{R}^{|\mathbb{J}_0|\times n}}{\arg\min}\ I(v;y), \tag{11}$$

where we recall that $I(v;y) = J(v) + \Phi(v;y)$ (see [7, pp. 66-67] for more details).

3 Discrete-Time Filtering Algorithms

We now briefly present some approximate algorithms for discrete-time filtering, where we stress once again that computing the posterior filtering distribution is in general unfeasible. We still mention a case where an explicit posterior distribution exists, which is given by the **Kalman Filter (KF)**, i.e. for the case where Ψ and h are linear. Thus we set $\Psi(v) = Mv$ and $h(v) = Hv$ such that $M \in \mathbb{R}^{n\times n}$ and $H \in \mathbb{R}^{m\times n}$. Denote by (m_j, C_j) the mean and covariance of $v_j|Y_j$ where we note that it completely characterizes the distribution thanks to Gaussianity. It is shown in [7, Theorem 4.1] that the update map of a KF is expressed in terms of a precision matrix (inverse of the covariance).

Theorem 3.1 (Kalman Filter)**.** *Assume $C_0, \Gamma, \Sigma \succ 0$. Then, $C_j \succ 0$ for all $j \in \mathbb{Z}^+$, and:*

$$C_{j+1}^{-1} = (MC_jM^T + \Sigma)^{-1} + H^T\Gamma^{-1}H, \tag{12}$$

$$C_{j+1}^{-1}m_{j+1} = (MC_jM^T + \Sigma)^{-1}Mm_j + H^T\Gamma^{-1}y_{j+1}, \tag{13}$$

where we further define $\widehat{m}_{j+1} := Mm_j$ and $\widehat{C}_{j+1} := MC_jM^T + \Sigma$. The later quantities represent the mean and covariance of $v_{j+1}|Y_j$.

Now, in the context of the variational method, we have that the mean of the KF satisfies:

$$m_{j+1} = \underset{v}{\arg\min}\ \frac{1}{2}|y_{j+1} - Hv|_\Gamma^2 + \frac{1}{2}|v - \widehat{m}_{j+1}|_{\widehat{C}_{j+1}}^2 =: \underset{v}{\arg\min}\ I_{\text{filter}}^{(j+1)}(v). \tag{14}$$

where $\widehat{m}_{j+1}$ and $\widehat{C}_{j+1}$ are defined as in Theorem (3.1). This relationship being for the fully Gaussian case, we can easily generalize the situation to non-linear dynamics:

$$m_{j+1} = \underset{v}{\arg\min}\ \frac{1}{2}|y_{j+1} - h(v)|_\Gamma^2 + \frac{1}{2}|v - \widehat{m}_{j+1}|_{\widehat{C}_{j+1}}^2 =: \underset{v}{\arg\min}\ I_{\text{filter},h}^{(j+1)}(v), \tag{15}$$

where h is now a non-linear function such that $\widehat{m}_{j+1} = \Psi(m_j) + \xi_j$ and with a yet to be defined $\widehat{C}_{j+1}$, which in this instance breaks Gaussianity. There are multiple models based on this construction, and we now mention a few of them:

- *3DVAR*: This model is constructed with $\xi_j \equiv 0$ and fixes a constant covariance $\widehat{C}_{j+1} \equiv \widehat{C}$, for all j.
- *Extended Kalman Filter* (ExKF): This model is constructed with $\xi_j \equiv 0$ for all j. It propagates the covariance according to the linearization of Eq. (1), $v_{j+1} = \Psi(v_j) + \xi_j$, and propagates the mean with the deterministic dynamics $v_{j+1} = \Psi(v_j)$.

We furthermore briefly describe two other important filters that require a slightly different setup.

3.1 Ensemble Kalman Filter (EnKF)

The Ensemble Kalman Filter generalizes the idea of approximate Gaussian filter, where rather than using the minimization procedure from Eq. (15) to update a single estimate of the mean, it is used to generate an *ensemble* of particles all of which satisfy the model inherent to the minimization procedure. The corresponding mean and covariance are then estimated using the ensemble, thereby adding further coupling to the particles in addition to that introduced by the data[7]. One possibility for such an algorithm is given by the *perturbed observation EnKF*:

$$\text{Prediction} \begin{cases} \widehat{v}_{j+1}^{(n)} = \Psi(v_j^{(n)}) + \xi_j^{(n)}, \quad n \in \{1,\dots,N\} \text{ and } \xi^{(n)} \overset{\text{iid}}{\sim} \mathcal{N}(0,\Sigma) \\ \widehat{m}_{j+1} = \frac{1}{N} \sum_{n=1}^{N} \widehat{v}_{j+1}^{(n)} \\ \widehat{C}_{j+1} = \frac{1}{N-1} \sum_{n=1}^{N} \left(\widehat{v}_{j+1}^{(n)} - \widehat{m}_{j+1}\right)\left(\widehat{v}_{j+1}^{(n)} - \widehat{m}_{j+1}\right)^T \end{cases}$$

$$\text{Analysis} \begin{cases} S_{j+1} = H\widehat{C}_{j+1}H^T + \Gamma \\ K_{j+1} = \widehat{C}_{j+1}H^T S_{j+1}^{-1} \\ y_{j+1}^{(n)} = y_{j+1} + \eta_{j+1}^{(n)}, \quad n \in \{1,\dots,N\} \text{ and } \eta^{(n)} \overset{\text{iid}}{\sim} \mathcal{N}(0,\Gamma) \\ v_{j+1}^{(n)} = (\mathbb{I} - K_{j+1}H)\widehat{v}_{j+1}^{(n)} + K_{j+1}y_{j+1}^{(n)}, \quad n \in \{1,\dots,N\} \end{cases}$$

Remark that this filter yields the Kalman Filter in the linear case and in the limit of infinite ensemble. Also, we have that this process does not necessarily generate a Gaussian filter. Another possibility to generate the preceding algorithm is to introduce the following family of cost functions:

$$I_{\text{filter},n}^{(j+1)}(v) := \frac{1}{2}|y_{j+1}^{(n)} - Hv|_{\Gamma}^2 + \frac{1}{2}|v - \widehat{v}_{j+1}^{(n)}|_{\widehat{C}_{j+1}}^2. \tag{16}$$

The prediction step uses the fully non-linear dynamics to determine $\{\widehat{v}_{j+1}^{(n)}\}_{n=1}^N$. From the prediction, the analysis step then determines the ensemble $\{v_{j+1}^{(n)}\}_{n=1}^N$ by minimizing $I_{\text{filter},n}$ with $n \in \{1,\dots,N\}$. The algorithm overall provides the mapping:

$$\left\{v_j^{(n)}\right\}_{n=1}^N \longmapsto \left\{\widehat{v}_{j+1}^{(n)}\right\}_{n=1}^N \longmapsto \left\{v_{j+1}^{(n)}\right\}_{n=1}^N. \tag{17}$$

From this algorithm, it is then natural do define an approximate filtering distribution from the empirical Dirac measures:

$$\mu_j^N := \frac{1}{N}\sum_{n=1}^{N} \delta_{v_j^{(n)}} \simeq \mu_j =: \mathbb{P}(v_j|Y_j) \quad \text{and} \quad \widehat{\mu}_{j+1}^N := \frac{1}{N}\sum_{n=1}^{N} \delta_{\widehat{v}_{j+1}^{(n)}} \simeq \widehat{\mu}_{j+1} =: \mathbb{P}(v_{j+1}|Y_j). \tag{18}$$

We remark that although these yield Monte Carlo estimators for the measures, in general (except for linear problems) these estimators do **not** converge to the true distributions μ_j and $\widehat{\mu}_j$ as $N \to \infty$. We also mention another variant of the EnKF named *Ensemble Square-Root Kalman Filter*, which we choose to not describe in this review. However one can refer to [7, pp. 86-87] for the details.

3.2 Particle Filters

In this section, we will only describe the *basic* idea behind particle filters, as it represents a vast topic. Now, it turns out that any probability measure which has a density with respect to the Lebesgue measure can be approximated with a finite convex combination of Dirac probability measures. We have already encountered this idea in the previous sections, such as for the EnKF in Eq. (18) where the weights where all equal. Once again, the role of a **particle filter** is to approximate the true filtering distribution $\mathbb{P}(v_j|Y_j)$. We recall that we denote by μ_j (resp. $\widehat{\mu}_{j+1}$) the probability measure on $\mathbb{R}^n$ which corresponds to the density $\mathbb{P}(v_j|Y_j)$ (resp. $\mathbb{P}(v_{j+1}|Y_j)$). The high-level idea of a particle filter is to estimate these densities with an N-particle Dirac measure approximation of the form:

$$\mu_j^N := \sum_{n=1}^N w_j^{(n)} \delta_{v_j^{(n)}} \simeq \mu_j \quad \text{and} \quad \widehat{\mu}_{j+1}^N := \sum_{n=1}^N \widehat{w}_{j+1}^{(n)} \delta_{\widehat{v}_{j+1}^{(n)}} \simeq \widehat{\mu}_{j+1}, \tag{19}$$

such that the weights must satisfy $\sum_n w_j^{(n)} = 1$ and $\sum_n \widehat{w}_{j+1}^{(n)} = 1$, for all j. Thus, we have that the approximate distribution μ_j^N is completely defined by the weights $\{w_j^{(n)}\}_n$ and particle positions $\{v_j^{(n)}\}_n$, and similarly $\widehat{\mu}_{j+1}^N$ is exclusively defined via some weights $\{\widehat{w}_{j+1}^{(n)}\}_n$ and particle positions $\{\widehat{v}_{j+1}^{(n)}\}$. Hence, particle filters aim at defining mappings of the form:

$$\left\{v_j^{(n)}, w_j^{(n)}\right\}_{n=1}^N \longmapsto \left\{\widehat{v}_{j+1}^{(n)}, \widehat{w}_{j+1}^{(n)}\right\}_{n=1}^N \longmapsto \left\{v_{j+1}^{(n)}, w_{j+1}^{(n)}\right\}_{n=1}^N, \tag{20}$$

thus corresponding to the prediction and analysis steps of a general filtering algorithm. We do not derive explicit prediction and analysis steps for particle filters for brevity. However, [7] proposes a *simple* particle filter based on *Sequential Importance Sampling* along with some improved proposals for the filtering distribution, while also making some analyses on the large-time behaviour of filters.

4 Data Assimilation in Continuous-Time

To derive some continuous-time analogues of the previous methods for data assimilation, we consider the discrete-time models in Eqs. (1) and (2) in the case where, for $\tau > 0$ we consider: $\Psi(\cdot) \mapsto \mathbb{I} \cdot + \tau f(\cdot)$, $y_{j+1} \mapsto (z_{j+1} - z_j)/\tau$, $\Sigma \mapsto \tau\Sigma_0$ and $\Gamma \mapsto \tau^{-1}\Gamma_0$. The assumption on Ψ, namely that it is a small $\mathcal{O}(\tau)$ perturbation of the identity, and the scaling of the model noise Σ to have variance $\mathcal{O}(\tau)$ ensures that the underlying signal process behaves as a stochastic differential equation (SDE) in the limit $\tau \to 0$[7]. As a result, we will consider a dynamical model given by the SDE

$$\mathrm{d}v(t) = f(v(t))\,\mathrm{d}t + \sqrt{\Sigma_0} \cdot \mathrm{d}B_v(t) \tag{21}$$

with $v(0) \sim \mathcal{N}(m_0, C_0)$, and an observation process defined by

$$\mathrm{d}z(t) = h(v(t))\,\mathrm{d}t + \sqrt{\Gamma_0} \cdot \mathrm{d}B_z(t), \tag{22}$$

such that $z(0) = 0$. Here, B_v (resp. B_z) is an $\mathbb{R}^n$-valued (resp. $\mathbb{R}^m$-valued) standard Brownian motion, such that they are independent of each other and also of $v(0)$. Now, from the scaling given

in the beginning of this section, the dynamical model will define the **signal** $\{v_j\}_{j\in\mathbb{Z}^+}$ taken to be an Euler-Maruyama approximation of the SDE (21):

$$v_{j+1} = v_j + \tau f(v_j) + \sqrt{\tau\Sigma_0}\cdot\widetilde{\xi}_j, \tag{23}$$

with $v_0 \sim \mathcal{N}(m_0, C_0)$ and $\widetilde{\xi} = \{\widetilde{\xi}_j\}_{j\in\mathbb{Z}^+}$ an i.i.d. sequence determined by $\widetilde{\xi}_0 \sim \mathcal{N}(0,\mathbb{I})$ independent of v_0. Moreover, we one again observe a (possibly non-linear) function of the signal where the **observations** are taken to be an Euler-Maruyama discretization of the observation process (22):

$$z_{j+1} = z_j + \tau h(v_{j+1}) + \sqrt{\tau\Gamma_0}\cdot\widetilde{\eta}_{j+1}, \tag{24}$$

with $z_0 = 0$ and $\widetilde{\eta} = \{\widetilde{\eta}_j\}_{j\in\mathbb{Z}^+}$ an i.i.d. sequence determined by $\widetilde{\eta}_1 \sim \mathcal{N}(0,\mathbb{I})$. We remark several independence properties implying that $\widetilde{\xi}$, $\widetilde{\eta}$ and v_0 are mutually independent random variables. We also note that the scaling has not been chosen arbitrarily, rather, it is a setup in which the resulting stochastic dynamics exhibit interesting properties without being dominated by noise or becoming fully deterministic. Once again, we will consider the problem of **data assimilation** to determine information about the signal v given data z.

4.1 Smoothing Problem

We review smoothing methods, where the objective is to determine the conditional probability distribution $v|z$ upon the observation of the process in Eq. (22) over a time-interval $[0,T]$ that will be suitably discretized. Once again, we will use Bayes' formula to determine this smoothing distribution. To do so, a key ingredient will be the Girsanov formula which describes how to find the change of measure between two SDEs with the same diffusion coefficient. Now, we recall some facts from Stochastic Analysis. SDEs of the form $\mathrm{d}v(t) = g(v(t),t)\,\mathrm{d}t + \gamma(v(t),t)\,\mathrm{d}W(t)$, with $v(0) = u$, generates strong solutions in $C([0,T];\mathbb{R}^d)$ as long as the smooth vector-valued function $g:\mathbb{R}^d\times\mathbb{R}^+\to\mathbb{R}^d$ and the smooth matrix-valued function $\gamma:\mathbb{R}^d\times\mathbb{R}^+\to\mathbb{R}^{d\times m}$ satisfy Lipschitz and linear growth conditions, and also $\mathbb{E}|u|^2 < \infty$. Here, W is a standard Brownian motion and the SDE is a Markov process with diffusion matrix $\Gamma:\mathbb{R}^d\times\mathbb{R}^+\to\mathbb{R}^{d\times d}$ given by $\Gamma(v,t) := \gamma(v,t)\gamma(v,t)^T$. Moreover, SDEs that do not have the same diffusion coefficient or that start from different deterministic initial conditions generate mutually singular measures. That is, two measures μ and ν on a measurable space $(\mathcal{X},\cdot)$ are *mutually singular* if there exists two disjoint measurable sets A,B with $A\cup B = \mathcal{X}$ such that $\mu(B) = 0$ and $\nu(A) = 1$. If this is not the case, then two different SDEs generate measures that are absolutely continuous. Now, we will require the Girsanov formula, which provides an explicit expression for the Radon-Nikodym derivative between two such measures. Let us present the Girsanov formula.

Theorem 4.1 (Girsanov Theorem)**.** *Consider the SDE* $\mathrm{d}v(t) = g(v(t))\,\mathrm{d}t + \gamma(v(t))\,\mathrm{d}W(t)$ *with* $v(0) = v_0$ *and the drift-less SDE* $\mathrm{d}v(t) = \gamma(v(t))\,\mathrm{d}W(t)$ *also with* $v(0) = v_0$*. Assume that* $v\in\mathbb{R}^p$*, the standard Brownian motion* $W\in\mathbb{R}^p$*, that* $g:\mathbb{R}^p\to\mathbb{R}^p$ *and* $\gamma:\mathbb{R}^p\to\mathbb{R}^{p\times p}$ *are Lipschitz on bounded sets, and that* $\gamma(v)$ *is invertible for every* $v\in\mathbb{R}^p$*. Assume also that the two SDEs have strong solutions on* $[0,T]$ *that do not explode almost surely. Then, the measures on* χ *and* χ_0 *(drift-less) on* $L^2([0,T];\mathbb{R}^p)$ *generated by the two SDEs respectively are equivalent with Radon-Nikodym derivative:*

$$\frac{\mathrm{d}\chi}{\mathrm{d}\chi_0}(v) = \exp\left[-\int_0^T\left(\frac{1}{2}\left|\gamma(v)^{-1}g(v)\right|^2\mathrm{d}t - \left\langle\gamma(v)^{-1}g(v),\gamma(v)^{-1}\,\mathrm{d}v\right\rangle\right)\right]. \tag{25}$$

Theorem 4.2 (Bayes' formula for infinite-dimensional spaces)**.** *Let κ, ζ be two probability measures on $S \times T$, where $(S, \cdot)$ and $(T, \cdot)$ are measurable spaces. Denote by (x, y) an element of $S \times T$. Assume that ζ dominates κ, $\kappa \ll \zeta$, meaning that if $\zeta(A) = 0$ for some measurable set A then $\kappa(A) = 0$. Assume also that κ has Radon-Nikodym derivative ϕ with respect to ζ, so that:*

$$\kappa(\mathrm{d}x, \mathrm{d}y) = \phi(x, y)\zeta(\mathrm{d}x, \mathrm{d}y) := \frac{\mathrm{d}\kappa}{\mathrm{d}\zeta}(x, y)\zeta(\mathrm{d}x, \mathrm{d}y). \tag{26}$$

Assume further that the conditional distribution of $x|y$ with respect to the measure ζ (resp. with respect to the measure κ), denoted $\zeta^y(\mathrm{d}x)$ (resp. $\kappa^y(\mathrm{d}x)$), exists. We also impose $\kappa^y \ll \zeta^y$. The Radon-Nikodym derivative is given by:

$$\frac{\mathrm{d}\zeta^y}{\mathrm{d}\kappa^y}(x) = \frac{\phi(x, y)}{\int_S \phi(x, y)\,\mathrm{d}\zeta^y(x)} \mathbb{1}_{\left\{\int_S \phi(x,y)\,\mathrm{d}\zeta^y(x) > 0\right\}}, \quad \text{for all } y \in T. \tag{27}$$

If the indicator function is not satisfied, then Eq. (27) equals 1.

Then, we understand that if we want to evaluate the smoothing distribution, we must calculate the Radon-Nikodym derivative $\phi(x, y)$ which can be done thanks to the Girsanov formula. Here, $\phi(x, y)$ represents the likelihood function[4], and we thus recognise a form of Bayes' formula in the case of infinite-dimensional spaces. Let $v = \{v(t)\}_{t\in[0,T]} \in L^2([0, T]; \mathbb{R}^n)$ and $z = \{z(t)\}_{t\in[0,T]} \in L^2([0, T]; \mathbb{R}^m)$. The **smoothing problem** in the continuous case is to study the signal v given the accumulated data up to time T, namely z, more precisely, to study the random variable $v|z$ on $L^2([0, T]; \mathbb{R}^n)$[7].

Theorem 4.3. *Let μ_0 denote the prior measure on the random variable $v \in L^2([0, T]; \mathbb{R}^n)$ defined by Eq. (21), and let μ denote the posterior measure for the random variable $v|z \in L^2([0, T]; \mathbb{R}^n)$ with z given by Eq. (22). Then, if f, h are bounded Lipschitz functions, it follows that $\mu \ll \mu_0$ and*

$$\frac{\mathrm{d}\mu}{\mathrm{d}\mu_0}(v) \propto \exp(-\Xi_2(v, z)), \tag{28}$$

where Ξ_2 is defined as:

$$\Xi_2(v, z) := \frac{1}{2}\int_0^T |h(v(t))|^2_{\Gamma_0}\,\mathrm{d}t - \int_0^T \langle h(v(t)), \mathrm{d}z(t)\rangle_{\Gamma_0}. \tag{29}$$

For instance, if we let ν_0 be the measure on $L^2([0, T]; \mathbb{R}^n)$ generated by the SDE $\mathrm{d}v(t) = \sqrt{\Sigma_0}\,\mathrm{d}B_v(t)$ such that $v(0) \sim \mathcal{N}(m_0, C_0)$, then ν_0 is a Wiener measure convolved with a random Gaussian initial starting point. From the Girsanov Theorem, we know that:

$$\frac{\mathrm{d}\mu_0}{\mathrm{d}\nu_0}(v) \propto \exp(-\Xi_1(v)), \tag{30}$$

where Ξ_1 is defined as:

$$\Xi_1(v) := \frac{1}{2}\int_0^T |f(v(t))|^2_{\Gamma_0}\,\mathrm{d}t - \int_0^T \langle f(v(t)), \mathrm{d}v(t)\rangle_{\Sigma_0}. \tag{31}$$

As a result, using Theorem (4.3), we conclude that:

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu_0}(v) \propto \exp(-\Xi_1(v) - \Xi_2(v, z)). \tag{32}$$

This expression is useful in the application of MCMC algorithms designed specifically for sampling probability measures defined via their density with respect to a Gaussian[7]. There is also the case of deterministic dynamics, however we will not delve into the situation in this review (see [7, pp. 165] for more details).

4.2 Filtering Problem

We consider the dynamical model in Eq. (21) and the observation process in Eq. (22). Define the accumulated data up to time t by $z^t := \{z(s)\}_{s\in[0,t]}$. The **filtering problem** aims at finding a sequential update of the measure μ^t, as t increases, which represents the distribution of the random variable $v(t)|z^t$. Recall that, associated to an SDE of the form $\mathrm{d}v(t) = g(v(t),t)\,\mathrm{d}t + \gamma(v(t),t)\,\mathrm{d}W(t)$ with $v(0) = u$, we characterise this Markov process by a *generator* $\mathcal{L}$ whose action on a test function $\varphi : \mathbb{R}^d \to \mathbb{R}$ is $\mathcal{L}\varphi = g\cdot\nabla\varphi + \frac{1}{2}\Gamma : \nabla\nabla\varphi$ where $\Gamma = \gamma\gamma^T$ is the diffusion matrix associated to the process. Moreover, associated to the generator $\mathcal{L}$, we define the L^2-adjoint of $\mathcal{L}$ denoted by $\mathcal{L}^*$ whose action on a test function $\varphi : \mathbb{R}^d \to \mathbb{R}$ is $\mathcal{L}^*\varphi = -\nabla\cdot(g\varphi) + \frac{1}{2}\nabla\cdot\nabla\cdot(\Gamma\varphi)$. Importantly, a random variable $v(t)$ solving the above SDE, it has density $\rho(\cdot,t)$ with respect to the Lebesgue measure on $\mathbb{R}^d$ such that ρ solves the so-called **Fokker-Planck equation**:

$$\frac{\partial\rho}{\partial t} = \mathcal{L}^*\rho. \tag{33}$$

We note that this differential equation is in general difficult to solve. Now, applying the *adjoint generator* $\mathcal{L}^*$ to a test function in the context of the SDE (21), one has:

$$\mathcal{L}^*\varphi = \nabla\cdot\left(-f\varphi + \frac{1}{2}\Sigma_0\nabla\varphi\right). \tag{34}$$

We have the following theorem about μ^t, in which $\mathbb{E}_{\mu^t}$ denotes expectation with respect to μ^t (and hence integration with respect to the density $\rho(\cdot,t)$). It characterizes the density of the filtering distribution (respectively unnormalized density of the filtering distribution) as the solution of the Kusher–Stratonovich stochastic partial differential equation (SPDE)[7]. Before stating the theorem, recall that in the absence of data, the probability density function for v satisfying Eq. (21) is given by the Fokker–Planck equation (33). The following theorem shows how this equation should be modified to incorporate conditioning on observed data.

Theorem 4.4. *Assume that f is globally Lipschitz, h is linearly bounded, and that the measure μ^t governing the filtering problem for Eqs. (21) and (22) has Lebesgue density $\rho(\cdot,t) : \mathbb{R}^n \to \mathbb{R}^+$ for each fixed t. Then, ρ solves the following Kushner-Stratonovich equation*

$$\frac{\partial\rho}{\partial t} = \mathcal{L}^*\rho + \rho\left\langle h - \mathbb{E}_{\mu^t}(h), \frac{\mathrm{d}z(t)}{\mathrm{d}t} - \mathbb{E}_{\mu^t}(h)\right\rangle_{\Gamma_0}. \tag{35}$$

Here, $\mathrm{d}z(t)/\mathrm{d}t$ represents the process described in Eq. (22). Furthermore, we have that $\rho(v,t) = r(v,t)/\int_{\mathbb{R}^n} r(v,t)\,\mathrm{d}v$, where r solves the Zakai equation

$$\frac{\partial r}{\partial t} = \mathcal{L}^* r + r\left\langle h, \frac{\mathrm{d}z(t)}{\mathrm{d}t}\right\rangle_{\Gamma_0}. \tag{36}$$

Note[7] that the Zakai equation is a linear stochastic PDE, while the Kushner–Stratonovich equation is a non-linear stochastic PDE, since evaluation of the expectation $\mathbb{E}_{\mu^t}$ requires integration against ρ.

5 Continuous-Time Smoothing Algorithms

We briefly describe algorithms for continuous-time smoothing. As usual, the simplest model is the case where f and h are linear giving the Gaussian property to the model. In particular, for $L \in \mathbb{R}^{n\times n}$ and $H \in \mathbb{R}^{m\times n}$ consider the posterior distribution $v|y$ described by Eqs. (21) and (22) in the case where $f(v) = Lv$ and $h(v) = Hv$. This is the so-called **Kalman-Bucy smoother**, and by Theorem (4.3) with a Gaussian measure μ_0 on $L^2([0,T];\mathbb{R}^n)$ it yields a characterization of the smoothing distribution as a Radon-Nikodym derivative. One can also explicitly identify the mean and covariance of the resulting Gaussian measure μ, however we omit the statement in this review[5,6]. The authors in [7, Theorem 7.2] also provide the mean and covariance of the Kalman-Bucy smoother for a deterministic linear dynamical model. Moreover, the authors in [7] also present some MCMC methods to sample from the posterior distribution in continuous-time, which is a hard task since we want to perform sampling from an infinite-dimensional space. They briefly describe the *Random Walk Metropolis (RWM)* sampler in the deterministic setting, along with the *independence dynamics sampler* and *pCN method* in the stochastic setting. However, it turns out that these methods can be quite inefficient, and as a result one we may turn to *variational methods*. The idea is to start from Eq. (32) for the measure μ on $v|z$ governed by Eqs. (21) and (22), while also considering the Gaussian measure ν_0 generated by the SDE $\mathrm{d}v(t) = \sqrt{\Sigma_0}\,\mathrm{d}B_v(t)$ such that $v(0) \sim \mathcal{N}(m_0, C_0)$. It has a constant mean $m(t) = m_0$ for all t, and a covariance defined by the Wiener measure with variance Σ_0. This gives a Gaussian measure $\mathcal{N}(m,C)$ on $L^2([0,T];\mathbb{R}^n)$. Now, we need a way for a MAP estimation to translate well to infinite dimensions, where there is no Lebesgue measure. More precisely, we must find points that maximize the probability of small balls centred at that point, and then consider the limit in which the small ball has radius that tends to zero (see [7, Theorem 3.12] for more details). Assume for a moment that ν_0 has a Lebesgue density, then it should have a density proportional to $\exp(-\frac{1}{2}|L^{\frac{1}{2}}(v-m)|^2)$ where $L = C^{-1}$ is to be specified. Thus, from Eq. (32), the posterior measure μ has a Lebesgue density proportional to $\exp(-I(v;z))$ where:

$$I(v;z) := \frac{1}{2}\left|L^{\frac{1}{2}}(v-m)\right|^2 + \Xi_2(v;z) + \Xi_1(v). \tag{37}$$

This, in turn, suggests that points that maximize probability are those that minimize the functional $I(\cdot;z)$; minimization is over functions v such that $v-m$ is in the domain of $L^{\frac{1}{2}}$. This motivating argument does lead to an infinite-dimensional minimization problem, which is particularly hard to solve because of the dependence on Ξ_1 and Ξ_2 (see, for instance, [2] for more details). See [7, pp. 180] for a description of MAP estimators in the deterministic case.

6 Continuous-Time Filtering Algorithms

We end this review by providing various algorithms to determine the filtering distribution μ^t in continuous time, based on the model in Eq. (21) and obversions in Eq. (22). Recall that we denote by z^t the data $\{z(s)\}_{s\in[0,t]}$ accumulated up to time t, where we are interested in the sequential update of the probability measure μ^t governing $v(t)|z^t$ in time. Once again, we first turn to linear problems by briefly describing the **Kalman-Bucy filter**. Since the filtering distribution is Gaussian in this case, the distribution is entirely characterized by the mean and covariance where the algorithm comprises a system of differential equations for their evolution. In particular, although the Kushner-Stratonovich and Zakai equations do not admit closed-form solutions in general, they do admit explicit expressions for the linear case.

Theorem 6.1 (Kalman-Bucy filter). *For $L \in \mathbb{R}^{n\times n}$ and $H \in \mathbb{R}^{m\times n}$, the filtering distribution μ^t for posterior $v(t)|z^t$ in the case where $f(v) = Lv$ and $h(v) = Hv$ is Gaussian, where the mean m and covariance C satisfy the following Kalman-Bucy equations:*

$$\frac{\mathrm{d}m}{\mathrm{d}t} = Lm + CH^T\Gamma_0^{-1}\left(\frac{\mathrm{d}z(t)}{\mathrm{d}t} - Hm\right), \tag{38}$$

$$\frac{\mathrm{d}C}{\mathrm{d}t} = LC + CL^T + \Sigma_0 - CH^T\Gamma_0^{-1}HC, \tag{39}$$

where we set the initial conditions $m(0) = m_0$ and $C(0) = C_0$. Here, $\mathrm{d}z(t)/\mathrm{d}t$ denotes the process in Eq. (22).

We now briefly present some approximate algorithms for continuous-time filtering, where we stress once again that computing the posterior filtering distribution is in general unfeasible. The starting point occurs in the case where Eq. (15) admits $h(v) = Hv$, and so the quadratic minimization problem is explicitly solvable with $S_{j+1} = H\widehat{C}_{j+1}H^T + \Gamma$, a **Kalman gain** $K_{j+1} = \widehat{C}_{j+1}H^T S_{j+1}^{-1}$, and an updated mean given by $m_{j+1} = (\mathbb{I} - K_{j+1}H)\widehat{m}_{j+1} + K_{j+1}y_{j+1}$. This solution is based on the assumption $\mathbb{P}(v_{j+1}|Y_j) = \mathcal{N}(\Psi(m_j), \widehat{C}_{j+1})$, where in addition to the update mean, one can obtain the updated covariance via $C_{j+1} = (\mathbb{I} - K_{j+1}H)\widehat{C}_{j+1}$.

6.1 3DVAR

The algorithm 3DVAR concerns the problem of deterministic dynamics with $\Sigma = 0$ and linear observations with $h(v) = Hv$. This model is also constructed with $\xi_j \equiv 0$ and fixes a constant covariance $\widehat{C}_{j+1} \equiv \widehat{C}$, for all j.

Theorem 6.2 (3DVAR). *In the continuous-time limit $\tau \to 0$, the filtering distribution μ^t of the posterior $v(t)|z^t$ for 3DVAR is Gaussian, where the mean m and covariance C satisfy the following differential equations:*

$$\frac{\mathrm{d}m}{\mathrm{d}t} = f(m) + CH^T\Gamma_0^{-1}\left(\frac{\mathrm{d}z(t)}{\mathrm{d}t} - Hm\right), \tag{40}$$

$$\frac{\mathrm{d}C}{\mathrm{d}t} = 0\,, \tag{41}$$

where we set the initial conditions $m(0) = m_0$ and $C(0) = \widehat{C}$. Here, $\mathrm{d}z(t)/\mathrm{d}t$ denotes the process in Eq. (22).

6.2 Extended Kalman Filter (ExKF)

The 3DVAR algorithm imposes a fixed covariance on the model in the prediction step of the algorithm, implying also a fixed covariance in the analysis step. The extended Kalman filter attempts to improve on this by propagating the covariance in the prediction step according to the linearized dynamics. Indeed, this model is constructed with $\xi_j \equiv 0$ for all j. It propagates the covariance according to the linearization of Eq. (1), $v_{j+1} = \Psi(v_j) + \xi_j$, and propagates the mean with the deterministic dynamics $v_{j+1} = \Psi(v_j)$.

Theorem 6.3 (ExKF). *In the continuous-time limit $\tau \to 0$ with stochastic dynamics and linear observations $h(v) = Hv$, the filtering distribution μ^t of the posterior $v(t)|z^t$ for ExKF is Gaussian, where the mean m and covariance C satisfy the following differential equations:*

$$\frac{\mathrm{d}m}{\mathrm{d}t} = f(m) + CH^T\Gamma_0^{-1}\left(\frac{\mathrm{d}z(t)}{\mathrm{d}t} - Hm\right), \tag{42}$$

$$\frac{\mathrm{d}C}{\mathrm{d}t} = Df(m)C + C(Df(m))^T + \Sigma_0 - CH^T\Gamma_0^{-1}HC\ , \tag{43}$$

where we set the initial conditions $m(0) = m_0$ and $C(0) = C_0$. Here, $\mathrm{d}z(t)/\mathrm{d}t$ denotes the process in Eq. (22) and $Df(m)$ denotes the Jacobian matrix of f evaluated at m.

6.3 Ensemble Kalman Filter (EnKF)

Recall that the ensemble Kalman filter differs from the extended Kalman filter and 3DVAR in that instead of using an appropriate minimization procedure to update a single estimate of the mean, a minimization principle is used to generate an ensemble of particles all of which satisfy the model/data minimization principle. These are coupled through the empirical covariance used to weight the minimization. Thus in studying the EnKF in continuous time, it is natural to derive an SDE for each of the particles instead of deriving a single equation for the mean and the covariance[7].

Theorem 6.4 (EnKF). *In the continuous-time limit $\tau \to 0$ with stochastic dynamics and linear observations $h(v) = Hv$, the particles evolve according to the following set of coupled SDEs, for $n \in \{1, \dots, N\}$:*

$$\frac{\mathrm{d}z^{(n)}}{\mathrm{d}t} = Hv + \Gamma_0^{1/2}\left(\frac{\mathrm{d}W^{(n)}}{\mathrm{d}t} + \frac{\mathrm{d}B_z}{\mathrm{d}t}\right), \tag{44}$$

$$\frac{\mathrm{d}v^{(n)}}{\mathrm{d}t} = f(v^{(n)}) + C(v)H^T\Gamma_0^{-1}\left(\frac{\mathrm{d}z^{(n)}}{\mathrm{d}t} - Hv^{(n)}\right) + \Sigma_0^{1/2}\frac{\mathrm{d}B_v^{(n)}}{\mathrm{d}t}, \tag{45}$$

where $W^{(1)}, \dots, W^{(N)}, B_z, B_u$ are mutually independent standard Brownian motions. The associated mean $m(v)$ and covariance $C(v)$ are defined empirically from the particles $v = \{v^{(n)}\}_{n=1}^N$ with the estimators:

$$m(v) = \frac{1}{N}\sum_{n=1}^{N} v^{(n)}, \tag{46}$$

$$C(v) = \frac{1}{N-1}\sum_{n=1}^{N}\left(v^{(n)} - m\right)\left(v^{(n)} - m\right)^T. \tag{47}$$

We will not delve into continuous-time particle filters, however the reader may refer to [1] and [3] for detailed accounts of the topic. We also mention that the authors in [7, pp.194-196] investigate the large-time behaviour of these continuous-time filters.

Notation

- $L^2([0,T];\mathbb{R}^k)$ is the Hilbert space of functions $f:[0,T]\to\mathbb{R}^k$ such that $\int_0^T \|f(t)\|_{\mathbb{R}^k}^2 \,\mathrm{d}t < \infty$.
- $\langle\cdot,\cdot\rangle$ and $|\cdot|$ are the Euclidean inner product and norm on $\mathbb{R}^k$ (occasionally extended to $L^2([0,T];\mathbb{R}^k)$).
- $\langle\cdot,\cdot\rangle_A = \left\langle A^{-\frac{1}{2}}\cdot, A^{-\frac{1}{2}}\cdot\right\rangle$ is the weighted inner product (A is a positive definite matrix).
- $|\cdot|_A = \left|A^{-\frac{1}{2}}\cdot\right|$ is the A-induced norm (A is a positive definite matrix).
- $A \succ 0$ (resp. $A \succeq 0$) defines a positive (resp. semi-positive) definite real symmetric matrix.
- $(A:B)$ is the Frobenius inner product of A and B with $(A:B) := \sum_{i,j} A_{ij}B_{ij} = \mathrm{Tr}(AB^T) = \mathrm{Tr}(BA^T)$.
- $\mathcal{N}(m,C)$ is a Gaussian probability distribution with mean m and covariance C on $\mathbb{R}^k$ or $L^2([0,T];\mathbb{R}^k)$.
- B is a standard Brownian motion entering the signal model (in continuous-time).
- W is a standard Brownian motion entering the data model (in continuous-time).
- Σ_0, Γ_0 are the covariances of B and W respectively.
- Σ, Γ are the covariances of ξ and η respectively.

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